MATH 241: Calculus III

Fall 2026

Lecture 5: Vector-Valued Functions

Background & Motivation

We now shift from static geometry to motion. A vector-valued function $\mathbf{r}(t)$ traces a curve through space as the parameter t varies—think of it as the position of a particle at time t. This chapter connects the vector algebra from Chapter 1 to calculus: we'll differentiate and integrate these functions component by component, leading to tangent vectors, arc length, curvature, and projectile motion.

1. Parametric Curves in $\mathbb{R}^3$

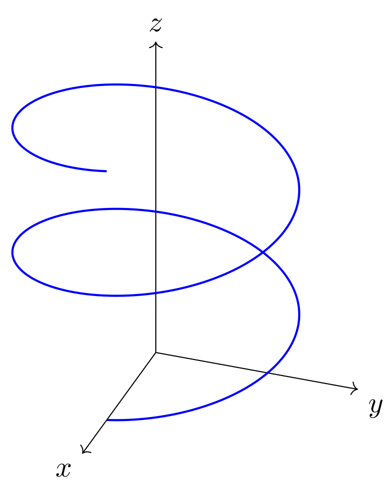
Definition 1: Vector-Valued Function (VVF)

A VVF maps a scalar t to a vector:

$$\mathbf{r}(t) = \langle f(t),\ g(t),\ h(t)\rangle = f(t)\,\hat{\mathbf{i}} + g(t)\,\hat{\mathbf{j}} + h(t)\,\hat{\mathbf{k}}$$

– f, g, h are the **component functions**

– As t varies, the tip of $\mathbf{r}(t)$ traces a **space curve**



Helix $\mathbf{r}(t) = \langle R\cos t,\ R\sin t,\ ct\rangle$ spiraling around the z-axis. As t increases, $\mathbf{r}(t)$ rotates in the xy-plane while rising linearly in z.

Common Parametric Curves

- **Line:** $\mathbf{r}(t) = \mathbf{r}_0 + t\,\mathbf{d}$
- **Circle** (radius R, xy-plane): $\mathbf{r}(t) = \langle R\cos t,\ R\sin t,\ 0\rangle, \quad t \in [0, 2\pi]$
- **Helix** (radius R, rising rate c): $\mathbf{r}(t) = \langle R\cos t,\ R\sin t,\ ct\rangle$
- **Ellipse:** $\mathbf{r}(t) = \langle a\cos t,\ b\sin t,\ 0\rangle$

Example 1: Identifying a Helix

Describe the curve $\mathbf{r}(t) = \langle 3\cos t,\ 3\sin t,\ 2t\rangle$ for $t \geq 0$.
1. Check: $x^2 + y^2 = 9\cos^2 t + 9\sin^2 t = 9$ – lies on the cylinder $x^2 + y^2 = 9$.
2. The z-component $z = 2t$ increases linearly.
3. Result: a **helix** of radius 3, rising at rate 2 per radian, winding around the z-axis.

2. Derivatives of VVFs

Definition 2: Derivative of a VVF

The derivative $\mathbf{r}'(t)$ is the instantaneous rate of change of the position vector, obtained by differentiating each component separately. Geometrically it is the **velocity vector**: it points in the direction the curve is moving at that instant, and its magnitude is the speed at which the particle is traversing the curve.

$$\mathbf{r}'(t) = \left\langle f'(t),\ g'(t),\ h'(t)\right\rangle$$

– $\mathbf{r}'(t)$ is a **tangent vector** to the curve at $\mathbf{r}(t)$
– $\|\mathbf{r}'(t)\|$ is the **speed**

Derivative Rules for VVFs

- Scalar product rule: $\dfrac{d}{dt}[f(t)\,\mathbf{r}(t)] = f'(t)\,\mathbf{r}(t) + f(t)\,\mathbf{r}'(t)$
- Dot product rule: $\dfrac{d}{dt}[\mathbf{r}(t)\cdot\mathbf{s}(t)] = \mathbf{r}'(t)\cdot\mathbf{s}(t) + \mathbf{r}(t)\cdot\mathbf{s}'(t)$
- Cross product rule: $\dfrac{d}{dt}[\mathbf{r}(t)\times\mathbf{s}(t)] = \mathbf{r}'(t)\times\mathbf{s}(t) + \mathbf{r}(t)\times\mathbf{s}'(t)$
- Chain rule: $\dfrac{d}{dt}[\mathbf{r}(g(t))] = \mathbf{r}'(g(t))\cdot g'(t)$

Example 2: Derivative and Analysis at a Point

Let $\mathbf{r}(t) = \left\langle t^3,\ \sin t,\ e^t\right\rangle$. Find $\mathbf{r}'(t)$ and analyze at $t = \pi$.
1. Derivative: $\mathbf{r}'(t) = \left\langle 3t^2,\ \cos t,\ e^t\right\rangle$
2. At $t = \pi$:
   - Position: $\mathbf{r}(\pi) = \left\langle \pi^3,\ 0,\ e^\pi\right\rangle$
   - Tangent vector: $\mathbf{r}'(\pi) = \left\langle 3\pi^2,\ -1,\ e^\pi\right\rangle$
3. Component analysis: x-comp > 0 (increasing), y-comp < 0 (decreasing), z-comp > 0 (increasing)
4. Speed: $\|\mathbf{r}'(\pi)\| = \sqrt{9\pi^4 + 1 + e^{2\pi}}$

3. Integrals of VVFs

Definition 3: Integral of a VVF

Also **component-wise**:

$$\int \mathbf{r}(t)\,dt = \left\langle \int f(t)\,dt,\ \int g(t)\,dt,\ \int h(t)\,dt \right\rangle + \mathbf{C}$$

– $\mathbf{C} = \langle C_1, C_2, C_3\rangle$ is a **vector constant** of integration

$$\int_a^b \mathbf{r}(t)\,dt = \left\langle \int_a^b f(t)\,dt,\ \int_a^b g(t)\,dt,\ \int_a^b h(t)\,dt \right\rangle$$

Example 3: Definite Integral of a VVF

Evaluate $\displaystyle\int_0^1 \langle 3t^2,\ 2\sin t,\ e^{2t}\rangle\ dt.$

1. x-component: $\displaystyle\int_0^1 3t^2\,dt = \left[t^3\right]_0^1 = 1$
2. y-component: $\displaystyle\int_0^1 2\sin t\,dt = [-2\cos t]_0^1 = 2 - 2\cos 1$
3. z-component: $\displaystyle\int_0^1 e^{2t}\,dt = \left[\frac{e^{2t}}{2}\right]_0^1 = \frac{e^2-1}{2}$
4. Result: $\left\langle 1,\ 2-2\cos 1,\ \frac{e^2-1}{2}\right\rangle$

4. Tangent Lines and Curves on Surfaces

Tangent Line at $t = t_0$

$$\mathbf{L}(s) = \mathbf{r}(t_0) + s\,\mathbf{r}'(t_0)$$

– Point of tangency: $\mathbf{r}(t_0)$ – Direction: $\mathbf{r}'(t_0)$

– To show a curve lies on a surface: substitute the components into the surface equation and verify

Example 4: Tangent Line + Curve on a Surface

The curve $\mathbf{r}(t) = \langle e^t,\ e^{-t},\ t\rangle$. Show it lies on $xy = 1$ and find the tangent line at $t = 0$.

1. On the surface: $x \cdot y = e^t \cdot e^{-t} = e^0 = 1$ ✓ for all t.
2. At $t = 0$: $\mathbf{r}(0) = \langle 1, 1, 0\rangle$
3. Tangent vector: $\mathbf{r}'(t) = \langle e^t, -e^{-t}, 1\rangle \implies \mathbf{r}'(0) = \langle 1, -1, 1\rangle$
4. Tangent line: $\mathbf{L}(s) = \langle 1+s,\ 1-s,\ s\rangle$

Show all work for full credit.

Practice Problems

Problem 1. If $\mathbf{r}(t) = \langle t^2,\ \ln(t+1),\ \cos t\rangle$, find $\mathbf{r}'(t)$ and $\mathbf{r}'(0)$.

Answer: $\mathbf{r}'(t) = \left\langle 2t,\ \frac{1}{t+1},\ -\sin t \right\rangle, \quad \mathbf{r}'(0) = \langle 0, 1, 0\rangle$

Problem 2. Evaluate $\displaystyle\int_0^{\pi} \langle \cos t,\ \sin t,\ 1\rangle\, dt$.

Answer: $\langle 0,\ 2,\ \pi\rangle$

Problem 3. A particle has velocity $\mathbf{v}(t) = \langle 2t,\ 3,\ -1\rangle$ and $\mathbf{r}(0) = \langle 1, 0, 5\rangle$. Find $\mathbf{r}(t)$.

Answer: $\mathbf{r}(t) = \langle t^2 + 1,\ 3t,\ 5 - t\rangle$

Problem 4. Show that $\mathbf{r}(t) = \langle 2\cos t,\ 3\sin t,\ t\rangle$ lies on the elliptic cylinder $\frac{x^2}{4} + \frac{y^2}{9} = 1$. Find the tangent line at $t = \pi/2$.

Answer: tangent line $= \langle -2s,\ 3,\ \pi/2 + s\rangle$

Problem 5. Prove: if $\|\mathbf{r}(t)\| = c$ (constant), then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$ for all t.

Hint: differentiate $\mathbf{r}(t) \cdot \mathbf{r}(t) = c^2$.

$\frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{r}(t)] = \frac{d}{dt}[c^2] = 0$

By dot product rule: $\mathbf{r}'(t) \cdot \mathbf{r}(t) + \mathbf{r}(t) \cdot \mathbf{r}'(t) = 0 \implies 2\,\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0.$ $\square$

Interpretation: a curve on a sphere has velocity always perpendicular to the position vector.